

Topic Modeling using Language Models

Berker Keleşoğlu

Advisor: Suzan Üsküdarlı

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1. Introduction

Topic modeling is used to extract meaningful topics from text and group documents accordingly¹. Especially, large documents are particularly hard to group, which may be significant problem in real-world cases. Topic modeling addressess this by discovering the hidden patterns in text without manual grouping. Moreover, it handles this task through several different approaches. Classical methods such as Latent Dirichlet Allocation(LDA)[1] have been widely used for this purpose. In this research, I compare three different approaches in topic modeling. With the rising popularity Large Language Models, topic modeling – like every other modeling field - is also evolving, with newer approaches such as BERTopic [2] and TopicGPT [3] emerging.

However, following popularity alone may be a mistake, so comparing well-established techniques with LLM-based approaches is what motivated me to conduct this research.

The three approaches compared in this research represent different generations of topic modeling. LDA is a classical statistical method that treats each document as a probabilistic mixture of topics and produces word lists as output. BERTopic, introduced in 2022, represents the embedding-based generation and uses transformer-based sentence embeddings combined with clustering to capture semantic context that earlier methods could not. TopicGPT, presented at NAACL 2024, represents the most recent generation, in which Large Language Models are used directly through engineered prompts to generate human-readable topic labels instead of word lists. Comparing these three approaches under the same conditions provides insight into how the field has evolved from purely statistical methods toward semantic and prompt-based methods.

The research follows two main questions. First, how do these three approaches differ in terms of topic interpretability and output format? Second, what are the practical trade offs between them regarding processing time, scalability, and cost? To answer these questions, the three methods are applied to the 20 Newsgroups dataset under the same experimental conditions and their outputs are compared both quantitatively through training time and topic count, and qualitatively through the alignment between assigned topics and the true categories of the documents.

1.1. Structure

This research report consists of five different sections. Section 1 explains the plan that I followed through my research. Section 2 provides an introduction to the topic and the motivation behind the research. Section 3 defines the problem by discussing the limitations of classical topic modeling methods and need for new approaches. Section 4 presents solution through examples from code practice that I compared three different approaches to topic modeling. Section 5 concludes the report with the main finding, challenges and future works. Overall structure of the report is summarized in Figure 2.1

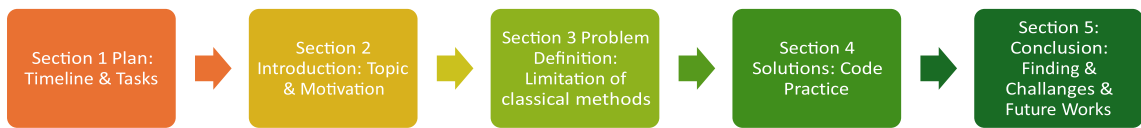


Figure 2.1: Structure of report

2. Background Information

Topic modeling was first studied back in the 1980s as a branch of generative modeling, which assumes that observed data is generated through a probabilistic relationship with hidden variables[4]. These early methods, often referred to as bag-of-words approaches, treated each document as an unordered collection of terms, which limited their ability to identify the underlying thematic structure of a text corpus. [4]. This is basically origin of Latent Semantic Analysis(LSA), introduced by Deerwester et.al in 1990[5]. LSA factorizes a term-document matrix using Singular Value Decomposition (SVD), placing conceptually related terms and documents close to one another in a reduced semantic space. This approach addressed two fundamental problems in information retrieval synonymy (different words referring to the same concept) and polysemy (a single word carrying multiple meanings) that simple keyword-based methods could not handle. Non-Negative Matrix Factorization(NMF) which is a different approach proposed by Lee and Seung in 1999[6], improved by interpretability by separating document-term matrix into non-negative factors, showing only collective combination, enabled us meaningful presentation. Because NMF restricts its factors to non-negative values, the resulting topics tend to correspond to interpretable groups of related words rather than abstract mathematical components.

A major turning point came with Latent Dirichlet Allocation(LDA), developed by Blei, A. Y. Ng, Jordan in 2003[1]. LDA takes each document as a mixture of topics and each topic as distribution over words. It was the most widely used topic modeling method and enabled benchmark for subsequent approaches. However, LDA has welldocumented limitations. It assumes the number of topcis is fixed and relies on bag of words assumption[4]. It is also sensitive to random initialization, which can produce different topics from the same data across multiple runs [4]. Many other extension was developed to solve these weaknesses, including correlated topic models for capturing

topic relationships, dynamic topic models for tracking how topics evolve over time, and pseudo-document approaches for handling short texts such as social media posts [4].

More recently, the rise of Large Language Models has shifted topic modeling from purely statistical methods towards semantic approaches. BERTopic uses transform based embeddings and class based TF-IDF method to capture semantic context that bag of words methods are not able to. It processes documents through a pipeline of embedding, dimensionality reduction with UMAP, clustering with HDBSCAN, and topic extraction replacing the word-frequency foundation of earlier methods with deep semantic understanding. TopicGPT takes this further by using LLM prompts directly to generate human readable topic labels.

Topic Quality Measurement

Evaluating topic modeling results are always challenging because topic quality can be change according to people perspective. The most common quantitative approach is topic coherence which compares semantic similarity between high score words inside topic. Ümass coherence is based on occurrence counts and it is fast. C_v coherence uses a sliding window combined with normalized pointwise mutual information and cosine similarity. C_{npmi} coherence scores -1 to 1 and range outputs according to score. For prompt based methods that produce natural language labels, measurement is totally based on human assessment or alignment with categories.

In addition to coherence based metrics two other evaluation approaches are commonly used. Topic diversity measures the proportion of unique words across topics and indicates whether a model produces distinct themes or repeats similar word groups. Traditionally used in language modeling measures how well a model predicts held-out documents and is often reported for LDA-based methods. However, these metrics are not directly comparable across the three approaches studied in this research, because LDA and BERTopic produce word-list topics while prompt based methods produce natural language labels. Thus the present research relies on qualitative analysis through same-category document assignments rather than a single numerical score.

3. Problem Definition

Main aim of topic modeling is discovering themes in large text collections automatically. This research compares three different approaches to solve this problem:

Latent Dirichlet Allocation(LDA) is a model that assumes every document is a mixture of a small number of hidden topics. Also it treats each topic as a distribution over words. This method is based on bag-of-words representation which is a way of representing text as a number so that algorithms can work with it and require number of topics to be specified in advance.

BERTopic is a modular topic modeling framework that converts documents into semantic embeddings. After clustering these embeddings, it extracts topic label using TF-IDF which is a method that calculates importance of a word in a document. Unlike LDA, it captures meaning of a word and context rather than only counting frequency.

Prompt-based approaches use Large Language Models to generate readable topic labels for documents. They include different prompt engineering techniques such as Persona, CoT and few-shot which improve the performance of LLMs when they are combined carefully. In this research, a combination of these prompt engineering techniques was implemented using Gemini 2.5 Flash Lite model, accessed through PydanticAI agentic development framework.

The problem is that despite availability of these approaches, their comparison in practice remains unclear. LDA and BERTopic both produce word-list outputs, but they differ from their techniques on word selection. However, prompt-based LLMs produce readable natural language labels instead of word lists. Each approach introduces different trade-offs in interpretability, time and cost. The main goal of this research is to compare these three different approaches under same conditions.

4. Solutions

4.1. Code Practice Setup

This section describes the setup which I have used during comparision the three topic modeling approaches under the same condition.

These practices use the 20 Newsgroup dataset which is generally used benchmark for text classificaiton and topic modeling. It includes 18,846 documents organized into 20 categories covering religion, sports, computer hardware, politics and science. The dataset is accessed through the scikit-learn library¹.

Compared Methods		
Approach	Method	Library / Model
Classical	LDA	scikit-learn
Embedding-based	BERTopic	BERTopic (sentencetransformers)
Prompt-based	LLM prompting	Gemini 2.5 Flash-Lite + PydanticAI

For LDA and BERTopic, dataset is fully used which consist of 18.846 documents. For prompt based approach, a sample of 200 documents(10 documents per category) was picked due to API rate limits and processing cost. This sampling ensures that every category is evaluated and represented in presentation equally.

The code is seperated into different modules for maintainbality and being able to see outputs clearly. A shared dataset module downloads the data and stores to prevent downloading again by different models. Each script produces both human-readable text file and machine-readbale JSON files.

¹ https://scikit-learn.org/stable/datasets/real_world.html#newsgroups-dataset

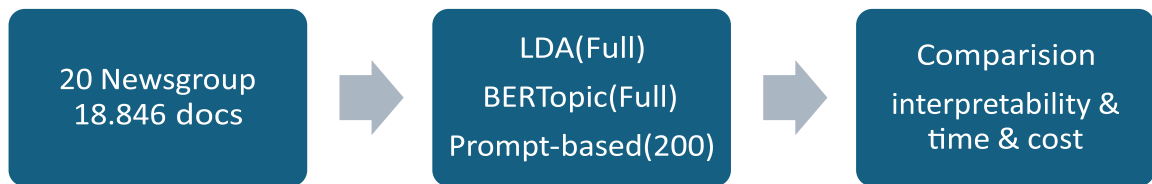


Figure 4.1 Overall Code Practice Setup

4.2. Implementation

Latent Dirichlet Allocation was implemented using scikit-learn. Documents were converted into document-term matrix by CountVectorizer with English stop-word removal by passing as parameter. The LDA model was fitted with 10 topics. The top 20 words from each topic were extracted as topic representation.

BERTopic approach didn't need much code because it handles embedding, clustering and topic extraction internally so these steps don't need to be configured manually. The model was fitted on full dataset and produced comparable number of topics. The top 20 words from each topic were extracted. Moreover, BERTopic generates an interactive visualization of the resulting topics.

Prompt-based approach implemented by using Gemini 2.5 Flash-Lite model and PydanticAI framework. A structured output model was defined so that the model returns topic label, a confidence score and three keywords for each document. System prompt combined three different prompt engineering techniques: a persona, Chain of Thought reasoning, few-shot examples. Some of the API calls returned errors due to API rate limits so a small number of the documents couldn't be sent to LLMs for topic extraction.

```
# output schema class
TopicResult(BaseModel):
    topic: str = Field(description="Short topic label (2-4 words)")
    confidence: float = Field(description="Confidence score between 0.0 and 1.0")
    keywords: list[str] = Field(description="Three key words from the text")
```


4.3. Results and Comparison

This part presents the result of three different approaches and compares them in terms of output format, topic quality, processing time and scalability.

Performance Comparison

Metric	LDA	BERTopic	Prompt-based
Documents processed	18.846	18.846	200(sample)
Number of topics	10	9	per-document
Processing time	48.6s	263.2s	476.9s
Output format	Word list	Word list	Natural language label
Preprocessing	Manual (stop words)	Integrated	Not required
Monetary cost	Free	Free	It has costs
Words per topic	20	20	Label

Table 4.3: Performance comparison of the three approaches

True Category	LDA Topic	BERTopic Topic	Promptbased Label
soc.religion.christian	god,people,jesus,said,did,know, think,church,christ,mr,say,president, don,children,believe,time,man,going, come, bible(Topic 2)	people,dont,just,think,know,god, said,government,like,say,did,time, right,believe,does,way,key,law,use, (Topic 0) mr	"Christian Discussion Group"
sci.electronics	drive,dos,card,use,windows,thanks,disk, know,scsi,does,like,pc,mac,software,problem, new,need,hard,used,work(Topic 1)	(outlier, no topic)	"Amateur Radio Literature"
rec.autos	just,don,like,good,think,know,time,ve,really,going, car,right,got,way,ll,make,people,better,want,didn (Topic 0)	(outlier, no topic)	"Automotive Mailing List Setup"

rec.sport.baseball	just,don,like,good,think,know,time,ve,really,going, car,right,got,way,ll,make,people,better,want,didn (Topic 0)	game,team,games,year,players,season, play,win,league,teams,period,think,baseball, good, la, dont, just, like, time, hit (Topic 2)	"Baseball Player Analysis"
sci.med	just,don,like,good,think,know,time,ve,really,going, car,right,got,way,ll,make,people,better,want,didn (Topic 0)	<i>(outlier, no topic)</i>	"Allergy Shot Effectiveness"
talk.politics.guns	just,don,like,good,think,know,time,ve,really,going, car,right,got,way,ll,make,people,better,want,didn (Topic 0)	people,dont,just,think,know,god, said,government,like,say,did,time, right,believe,does,way,key,law,use, mr (Topic 0)	"Gun Control Legislation"

Table: How the same documents were classified by each approach

Topic Quality

To be able to evaluate three approaches fairly, same documents were examined across all methods using the topic which each document was assigned to. The table above shows representative examples.

LDA's output were unclear. Some were clearly interpretable in word lists but such as a religion document (soc.religion.christian) was assigned a unrelated topic dominated by "people,don,think,just". An automotive document (rec.autos) was placed in a topic about "file,edu,image,program" and a baseball document was assigned to the generic "just, don, like, good" topic. This shows that LDA's performance in document-to-topic assignments are often inaccurate. A key finding is that evaluating LDA only with topic word list is wrong, because the words may look coherent while the actual document assignments are not.

BERTopic produced more coherent and thematically focused topics than LDA. However, it frequently used outlier category. When BERTopic assigned a topic, it was usually relevant. For example, the baseball document was correctly placed in a "game, team, player, season" topic. Several categories, such as electronics and medicine, were marked entirely as outliers and received no topic. Although it marked several

documents as outliers, it increases the precision because it doesn't force every document into a topic.

Prompt based approach produced accurate and readable labels for each processed document. Each document match the category and label with no generic or noise assignment and no unclassified documents.

Discussion

The results reveal clear trade-off between interpretability and cost. LDA is fastest but it fails in document-to-topic assignment and it requires manual interpretation. BERTopic offers significantly better topic coherence in five times longer waiting time and it still requires word-list interpretation though. The prompt-based approach produces the most directly usable output but it is the slowest among these approaches. It has costs and couldn't be scaled to full dataset due to API constraints.

Beyond these surface-level differences the three methods also differ in how they handle uncertainty. LDA forces every document into one of its topics even when the document does not clearly belong to any of them which is why four unrelated categories collapsed into the same generic topic in this experiment. BERTopic takes the opposite approach by marking unclear documents as outliers rather than assigning them incorrectly which improves precision but reduces coverage. The prompt-based method handles uncertainty differently again expressing it through confidence scores attached to each generated label. This difference is not just technical it reflects how each method views the boundary between a topic and noise, and it directly affects how reliable the results are when applied to real-world data where document quality and clarity vary significantly.

No single method is universally superior. The choice must be done according to use-case: LDA for speed and large-scale exploration, BERTopic for coherent topics without manual interpretation cost and prompt-based methods when human readable labels are the priority and dataset is suitable and small.

5. Conclusion

This research compared three different approaches to topic modeling. LDA, BERTopic and Prompt-based approaches were compared on the 20 Newsgroups dataset. The result shows that there is a clear trade-off between speed, accuracy and interpretability. LDA was fastest but produced inaccurate output in document-to-topic assignment. BERTopic created better output than LDA but left some documents as unclassified outliers. The prompt-based approach produced the most accurate output. However it was the slowest and may have significant costs with huge datasets. Also, prompt-based may face some API constraints with some LLMs. Overall, no single method proved universally superior. The choice depends on specific use cases.

This research also produced a broader methodological result evaluating topic models only by their topic word lists can be misleading. LDA's word lists looked reasonable in isolation but the same-category analysis revealed that four semantically unrelated categories were all collapsed into a single generic topic. This finding shows that topic word lists and document assignments must be examined together to understand a method's true performance. Another result is that the evolution from statistical methods (LDA) to embedding-based methods (BERTopic) and finally to prompt-based methods (TopicGPT-style approaches) is not a linear improvement in which every newer method outperforms older ones. Each generation introduces new strengths but also new constraints, such as monetary cost, scalability limits, and dependence on external services. Understanding these trade-offs is more valuable than choosing a single "best" method, and this awareness will guide the selection of topic modeling techniques in future work.

5.1. Challenges

BERTopic is sensitive to preprocessing because stop-word filtering significantly affected topic quality. API rate limits prevented the prompt-based method from processing full dataset. A further challenge was that the three methods produce different output formats.

5.2. Open problems and future work

The prompt-based approach could be tested with other LLMs to compare how different models perform on the same task. Also, additional classical methods such as LSA and NMF could be included into research and outputs can be evaluated in broader environment. As a last, prompt-based method can be applied to full dataset rather than a sample, it will allow cost analysis of LLM based topic modeling. Based on the sample results, processing all 18,846 documents would cost roughly \$7 to \$20 depending on the model used providing a clear picture of real-world scalability. A further extension would be exploring hybrid approaches that combine BERTopic's clustering with LLM-based labeling, potentially offering both the scalability of embedding methods and the readability of prompt-based methods.

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